

Definition 1 (Exact excitation matrices). For $\Omega \in \mathbb{C}^A$ and $B \subseteq \{1, \dots, k\}$, set

$$R_B = \bigotimes_{j=1}^k T_j, \quad T_j = \begin{cases} I - |\Omega\rangle\langle\Omega|, & j \in B, \\ |\Omega\rangle\langle\Omega|, & j \notin B. \end{cases}$$

If $\|\Omega\| = 1$, then each R_B is an orthogonal projection. The empty tensor product is the identity on $\mathbb{C}$.

1 Symmetry of exact excitation components

Let $\mathcal{H} = \mathbb{C}^A$, let $\Omega \in \mathcal{H}$, and write R_B for the exact excitation matrix indexed by $B \subseteq \{1, \dots, k\}$. When Ω is unit, this is the orthogonal projection of Definition 1. Let U_σ permute the physical copies by $\sigma \in S_k$. The following identities supply the component symmetries used in [1, Section 7].

Theorem 1 (Covariance of exact excitation sectors). For every B and σ , $U_\sigma R_B = R_{\sigma B} U_\sigma$.

Proof. Each matrix entry of R_B is the product of one-copy entries of $I - |\Omega\rangle\langle\Omega|$ at the indices in B and of $|\Omega\rangle\langle\Omega|$ at the remaining indices. Reindexing this product by σ^{-1} gives $R_{\sigma B}(\sigma x, \sigma y) = R_B(x, y)$, which is the stated covariance. $\square$

Theorem 2 (Stabilizer symmetry with auxiliary systems). If $\sigma B = B$, then $R_B \otimes I_K$ commutes with $U_\sigma \otimes M$ for every operator M on any finite-dimensional auxiliary space $\mathcal{K}$.

Proof. Covariance gives $U_\sigma R_B = R_B U_\sigma$. The two tensor products therefore multiply in either order to $(R_B U_\sigma) \otimes M$. $\square$

Theorem 3 (Symmetry inherited by an excitation component). Suppose $\sigma B = B$ and $(U_\sigma \otimes M)v = v$. Then the component $v_B = (R_B \otimes I_K)v$ satisfies $(U_\sigma \otimes M)v_B = v_B$.

Proof. Commute the two operators and apply the assumed fixed-vector identity. $\square$

For a simultaneously symmetric vector, this proves separate symmetry under permutations within B and within its complement. Taking σ to be the identity and M to be an auxiliary label projector preserves its given label. All three identities include $k = 0$ and require no normalization of Ω .

References

- [1] OpenAI, *A two-dimensional area law from a global spectral gap*, September 24, 2026.